

Geometric Representation of Amplitude Amplification in Grover's Algorithm

Optional, advanced. This note goes with Lab 3.5 (Grover's algorithm) and is not needed to complete the lab. It assumes the bra-ket notation of the labs: the inner product $\langle a|b\rangle$ of two states, and the outer product $|w\rangle\langle w|$, which acts on a state as $|w\rangle\langle w|\psi\rangle = \langle w|\psi\rangle |w\rangle$. It also uses sines and cosines of an angle and the idea of reflecting a vector across a line in a plane.

The Geometric Picture

Grover's algorithm operates in a two-dimensional subspace of the full N -dimensional Hilbert space ($N = 2^n$ for n qubits). The key insight is that the algorithm's state vector always remains in the plane that contains two vectors:

- $|s\rangle$: the uniform superposition

$$|s\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle, \quad N = 2^n.$$

- $|w\rangle$: the target (marked) state.

These two vectors are *not* orthogonal: $\langle w|s\rangle = 1/\sqrt{N}$. Define instead the normalized sum of all the other basis states,

$$|s'\rangle = \frac{1}{\sqrt{N-1}} \sum_{x \neq w} |x\rangle,$$

which is orthogonal to $|w\rangle$. The pair $\{|w\rangle, |s'\rangle\}$ is an orthonormal basis of the plane (not of the full space, which has N dimensions). All the calculations below use this basis.

Initial State and Angle θ

The algorithm starts in $|s\rangle$. We can write

$$|s\rangle = \sin \theta |w\rangle + \cos \theta |s'\rangle,$$

where

$$\sin \theta = \frac{1}{\sqrt{N}}, \quad \cos \theta = \sqrt{\frac{N-1}{N}}, \quad \theta = \arcsin\left(\frac{1}{\sqrt{N}}\right).$$

So θ is the angle between $|s\rangle$ and the $|s'\rangle$ axis. For large N , $\theta \approx \frac{1}{\sqrt{N}}$ is small. For $N = 8$, $\theta \approx 0.3614$ rad $\approx 20.7^\circ$.

Grover Iteration as Reflections

A single Grover iteration $G = DO$ consists of two reflections:

1. Oracle Reflection O

The oracle reflects about the hyperplane orthogonal to $|w\rangle$:

$$O = I - 2|w\rangle\langle w|.$$

It flips the sign of the $|w\rangle$ component and leaves every state orthogonal to $|w\rangle$ unchanged. Geometrically, O is a reflection across the line $|s'\rangle$ in our plane.

2. Diffusion Reflection D

The diffusion operator reflects about $|s\rangle$:

$$D = 2|s\rangle\langle s| - I.$$

Composition: Rotation by 2θ

The product of two reflections in a plane is a rotation by twice the angle between their axes. Here the two axes are $|s'\rangle$ (for O) and $|s\rangle$ (for D), at angle θ , so DO rotates the state vector by angle 2θ toward $|w\rangle$:

$$|s\rangle \xrightarrow{G} |\psi_1\rangle \xrightarrow{G} |\psi_2\rangle \xrightarrow{G} \dots \xrightarrow{G} |\psi_R\rangle \approx |w\rangle.$$

After $R \approx \frac{\pi}{4}\sqrt{N}$ iterations, the state is nearly $|w\rangle$.

Geometric Diagram

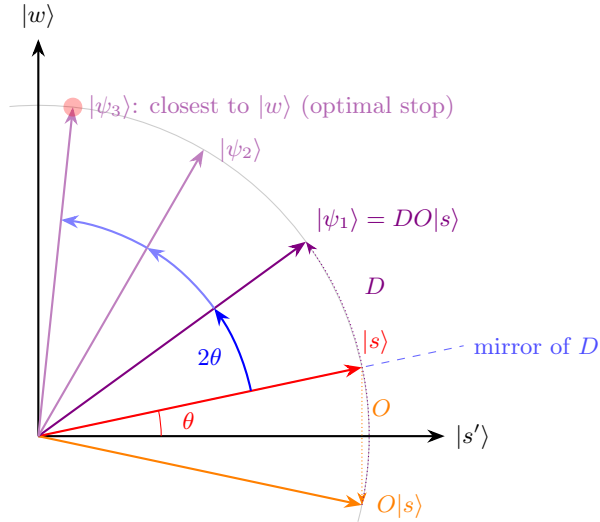


Figure 1: Geometric picture of Grover's algorithm in the plane with orthonormal axes $|s'\rangle$ and $|w\rangle$; all states are drawn from the origin and have length 1. The oracle O reflects $|s\rangle$ across the $|s'\rangle$ axis, to angle $-\theta$; the diffusion D reflects the result across the $|s\rangle$ line, to angle 3θ . Each iteration $G = DO$ therefore rotates the state toward $|w\rangle$ by 2θ . *Schematic:* the figure uses $\theta = 12^\circ$ for readability, so the closest approach to $|w\rangle$ is after 3 iterations; for $N = 8$, $\theta \approx 20.7^\circ$ and the best stop is after 2 iterations.

Mathematical Derivation

State After One Iteration

Start with $|s\rangle = \sin \theta |w\rangle + \cos \theta |s'\rangle$.

Oracle reflection:

$$O|s\rangle = \sin\theta O|w\rangle + \cos\theta O|s'\rangle = -\sin\theta|w\rangle + \cos\theta|s'\rangle,$$

which is $|s\rangle$ reflected across the $|s'\rangle$ axis: it makes angle $-\theta$ with $|s'\rangle$, and angle $\frac{\pi}{2} + \theta$ with $|w\rangle$ (its overlap with $|w\rangle$ is $-\sin\theta = \cos(\frac{\pi}{2} + \theta)$).

Diffusion reflection: Since D reflects about $|s\rangle$,

$$D|\psi^-\rangle = 2\langle s|\psi^-\rangle|s\rangle - |\psi^-\rangle,$$

where $|\psi^-\rangle = O|s\rangle$. Compute

$$\langle s|\psi^-\rangle = \sin\theta(-\sin\theta) + \cos\theta(\cos\theta) = \cos(2\theta).$$

Thus

$$D|\psi^-\rangle = 2\cos(2\theta)|s\rangle - |\psi^-\rangle.$$

In the $\{|w\rangle, |s'\rangle\}$ basis, the $|w\rangle$ component is $2\cos(2\theta)\sin\theta + \sin\theta = \sin(3\theta)$ and the $|s'\rangle$ component is $2\cos(2\theta)\cos\theta - \cos\theta = \cos(3\theta)$ (using $2\cos 2\theta \sin\theta = \sin 3\theta - \sin\theta$ and $2\cos 2\theta \cos\theta = \cos 3\theta + \cos\theta$). So

$$|\psi_1\rangle = \sin(3\theta)|w\rangle + \cos(3\theta)|s'\rangle.$$

The angle with $|s'\rangle$ increased from θ to 3θ —a rotation by 2θ .

General Iteration

After k iterations,

$$|\psi_k\rangle = \sin((2k+1)\theta)|w\rangle + \cos((2k+1)\theta)|s'\rangle.$$

The probability of measuring $|w\rangle$ is

$$P_k = \sin^2((2k+1)\theta).$$

P_k reaches its first peak when the angle $(2k+1)\theta$ is closest to $\frac{\pi}{2}$, that is for k close to $\frac{\pi}{4\theta} - \frac{1}{2}$. The standard choice is

$$k_{\text{opt}} = \left\lfloor \frac{\pi}{4\theta} \right\rfloor \approx \frac{\pi}{4} \sqrt{N},$$

for which $(2k_{\text{opt}}+1)\theta$ is within θ of $\frac{\pi}{2}$, so $P_{k_{\text{opt}}} \geq \cos^2\theta = 1 - \frac{1}{N}$. For $N = 8$: $\frac{\pi}{4\theta} \approx 2.17$, $k_{\text{opt}} = 2$ and $P_2 = \sin^2(5\theta) \approx 0.945$. More iterations do not help: the rotation continues past $|w\rangle$, and $P_3 = \sin^2(7\theta) \approx 0.330$.

Amplitude Amplification Generalization

The same geometric picture underlies *amplitude amplification*, a generalization where the initial state $|s\rangle$ has a small overlap with a “good” subspace $\mathcal{G}$ spanned by several basis states. Let $\Pi_{\mathcal{G}} = \sum_{x \in \mathcal{G}} |x\rangle\langle x|$ be the projector onto $\mathcal{G}$, and $\sin\theta$ the length of $\Pi_{\mathcal{G}}|s\rangle$. The oracle becomes $O = I - 2\Pi_{\mathcal{G}}$ (it flips the sign of every good component), and $D = 2|s\rangle\langle s| - I$ still reflects about the prepared state. The picture above then holds with $|w\rangle$ replaced by the normalized good part $\Pi_{\mathcal{G}}|s\rangle / \sin\theta$: each iteration rotates by 2θ , and the projection onto $\mathcal{G}$ grows until its first peak.

This framework extends Grover’s algorithm to quantum counting, collision finding, and many optimization problems.

Key Insights

- Grover's algorithm is a *rotation* in a 2D subspace by fixed angle 2θ per iteration.
- The oracle O “marks” the target by phase flip (reflection).
- The diffusion D amplifies via inversion about the mean (second reflection).
- Optimal iterations $\sim \frac{\pi}{4}\sqrt{N}$ yield success probability at least $1 - \frac{1}{N}$; stopping later lowers it again.
- Quadratic speedup in the number of oracle queries for unstructured search: $O(\sqrt{N})$ vs. classical $O(N)$.